

AI Chatbot Using Python - Project Synopsis

Project Title

AI Chatbot Using Python

Introduction

This project develops a simple AI chatbot using Python. The chatbot interacts with users through the command line, recognizes predefined keywords, provides suitable responses, and stores chat history with timestamps.

Objectives

- Build a chatbot using Python.
- Learn dictionaries, loops, functions, and file handling.
- Save conversations for future reference.
- Demonstrate basic keyword-based NLP.

Technologies Used

Python 3.x, VS Code, datetime module, text file storage.

Features

Interactive chatbot, predefined responses, keyword matching, chat history logging, timestamp recording, continuous conversation until 'bye'.

Working

The program displays a welcome message, accepts user input, converts it to lowercase, searches for matching keywords in a dictionary, prints the corresponding response, saves the conversation in chat_history.txt with date and time, and exits when the user types 'bye'.

Advantages

Simple to understand, demonstrates core Python concepts, easy to extend with GUI or AI APIs, suitable as a beginner portfolio project.

Limitations

Supports only predefined responses and does not understand complex natural language.

Future Enhancements

Add Tkinter GUI, voice recognition, OpenAI/Gemini API integration, database storage, and user authentication.

Conclusion

This project provides a strong foundation in chatbot development and demonstrates essential Python programming concepts useful for AI and software development.